

CSE6364 Software Evolution and Maintenance

Tutorial 8

Topic (Lecture 8): Software Change Management 1

1. Describe the external and internal sources of change for a software that is under maintenance.

External changes are driven by requirements and forces originating from outside the software itself to adapt to its environment. A primary driver is meeting new customer needs or responding to shifting new market conditions. Changes are also heavily triggered by modifications in the software's operating environment, which includes adapting to updated government policies or business rules. Finally, the introduction of new equipment or changes to the underlying software and hardware operating platforms forces external modifications to ensure the system remains compatible and fully functional.

Internal changes originate from within the system or organization to fix issues, optimize performance, or address the structural health of the software. The most critical internal driver is the requirement to repair errors and defects, which include design errors, logic errors, coding errors, and data processing flaws. Organizations also initiate internal changes to address performance or reliability issues or to expand existing requirements by enhancing functionality and improving computational efficiency. Lastly, internal changes are driven by organizational constraints, such as business reorganization or budgetary and scheduling constraints, as well as preventive work like code restructuring and optimization to combat a deteriorating software structure.

2. Why is it important to identify the type of changes when maintaining a software?
In principle, software maintenance activities can be classified individually. In practice, they are usually intertwined. E.g. in the course of modifying a program due to the introduction of a new operating system (adaptive change), obscure bugs may be introduced. The bugs have to be traced and dealt with corrective maintenance. Similarly, the introduction of a more efficient sorting algorithm into a data processing package (perfective maintenance) may require that the existing program code be restructured (preventive maintenance).
3. What is the main purpose of software change management?
The main purpose of software change management is to keep track with the changes that are requested by users, developers, or market forces, and ensuring that they are implemented in the most cost-effective way.
4. Describe the key activities in a software change control process.

- **Initiation and Impact Analysis:** The process begins when software changes, enhancements, or problems are identified. The maintenance team then analyzes the request to assess its impact, which helps the team understand the possible implications, identify affected files/documents, and estimate the necessary effort.
- **Proposal Preparation:** The results and compiled data from the impact analysis are gathered into a document known as a Change Proposal. This proposal outlines additional details, such as the resources needed and a proposed timeline for implementation.
- **Evaluation and Review:** The Change Control Board (CCB)—consisting of major stakeholders, project/operations teams, and technical experts—evaluates the change proposal. They assess the request based on its priority, impact analysis, budget, schedule, and potential disruption to productivity.
- **Decision and Disposition:** The board makes a final decision on the proposal. If rejected, the change is archived, and the originator is notified. If the change is not urgent or requires more details, the decision can be deferred. If approved, the change moves forward to execution.
- **Implementation and Verification:** Approved changes are incorporated through a formal process that starts with requirements analysis and updating, followed by standard software development activities. The code modifications are thoroughly tested, and all associated system documentation is reviewed and updated to verify correctness before releasing the software.
- **Feedback:** After the changes are incorporated and verified (or archived), feedback is officially supplied back to the originator to close out the change control loop.

5. What is impact analysis and why is it important in the software change control process?

Impact analysis is a core activity in the change control process that provides an accurate understanding of the implications, affected elements (such as source code, databases, and test cases), and estimated effort/tasks required to implement a proposed software change. It is critical to the software maintenance lifecycle because it helps the team and the Change Control Board (CCB) make informed business decisions about which proposals to approve, enables proper prioritization of changes based on relative benefits and technical risks, and forms the foundational basis for the formal Change Proposal document. Furthermore, conducting a thorough impact analysis helps prevent unforeseen "ripple effects," where an unstudied modification to one part of a program unpredictably distorts the logic and degrades the overall structure of other sections.

6. What is the role of the Change Control Board? Discuss who could be the members of this board.

CBB is a committee that makes decisions regarding whether or not proposed changes to a software should be implemented. The members may include representatives from major stakeholders, project, and operations team. Typically the board would include Subject Matter Experts and Technical Chiefs who can assess key technical and operational aspects that would be impacted by the decision.